# The Visual System

## The hierarchy of cortical visual processing

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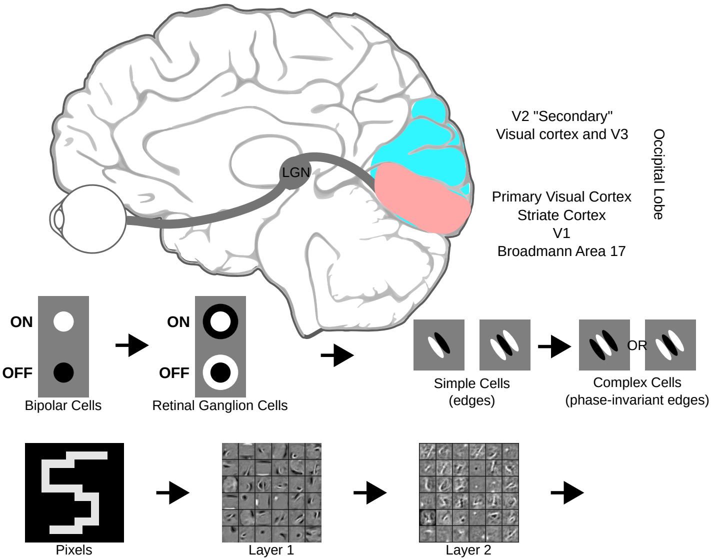
mrule

**Learning outcomes:**

- Describe the transformation of visual input from the retina to V1 and onward into more abstract cortical areas
- Describe how receptive fields and tuning properties change with increasing depth into the cortical visual processing hierarchy
- Briefly describe the distinction between “what” and “where” pathways in visual processing

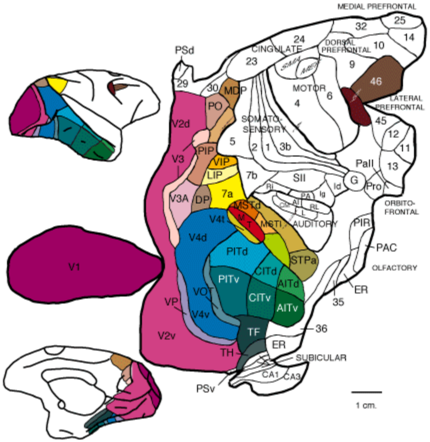
*The hierarchy of cortical visual processing*

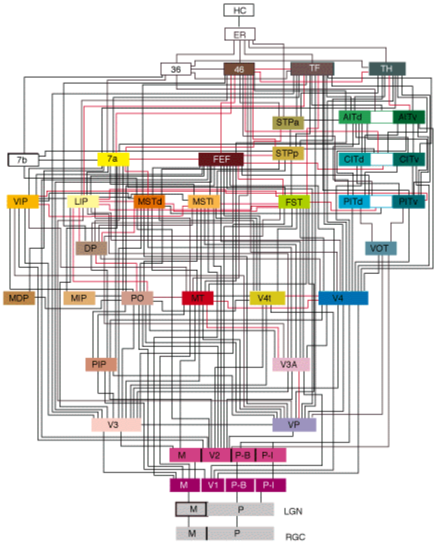
Previous slides reviewed the transformation of visual input from retinal on/off cells through V1 complex cells




neural network receptive fields from Erhan et al. (2009)

Now…





Felleman and van Essen, Cerebral Cortex (1991)

# *“What” vs “Where”*

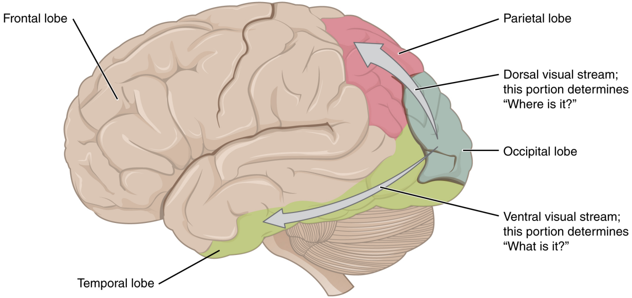



*illustration by OpenStax College*

***Two-streams hypothesis*** Primates split visual processing into 2 specialised pathways

**ventral stream, *‘what’*** object identity, semantics

- temporal lobe, language & memory

**dorsal stream, *‘where’*** location, geometry, speed

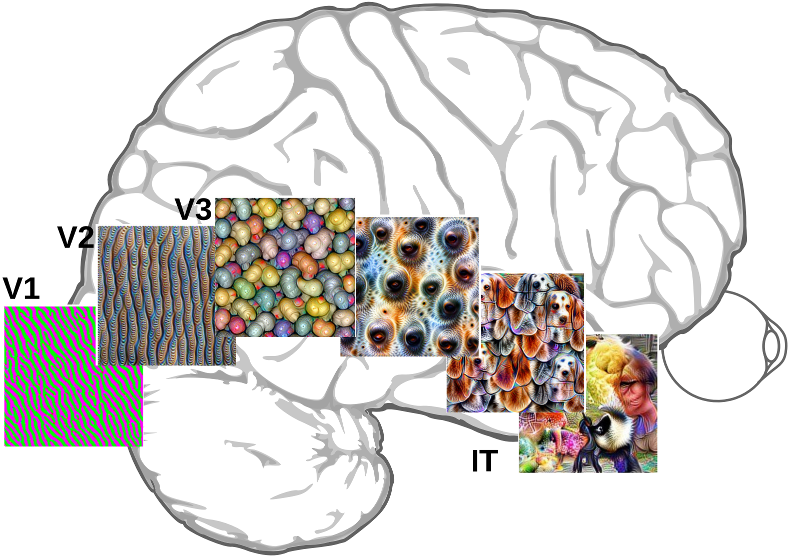
- parietal lobe ↔ motor cortex

# Coding along the visual hierarchy

Receptive fields change across visual-processing hierarchy

With increasing depth, receptive fields become:

- Larger
  - Pool together larger regions of visual input
  - More invariant to the specific location of an input
- Sensitive to more complicated aspects of the visual stimulus
  - e.g. recognising specific objects, or certain complicated motions
- More multi-modal (i.e. also depend on non-visual sensory signals)
  - Multi-sensory integration in parietal cortex
- More sensitive to top-down, contextual information
  - Task context, animal's behavioural state, attention, etc.




neural network receptive fields OpenAI microscope via Götz-Hahn et al. (2023)

# *The hierarchy of cortical visual processing*

Neurons sensitive to an object's movement, irrespective of its identity, are more likely to be found in the _____ or "where" cortical processing stream.

The ___________ ventral ("what") processing stream is responsible for recognising, naming, and remembering visually encountered objects based on their identity.

Receptive fields become (more/less) ____ spatially invariant with increasing depth into the cortical visual processing hierarchy

Would you expect V1 to be (more/less) ____ amenable to top-down modulation compared to area IT, deep in the ventral processing stream?

A cortical representation of an object's position, shape, and relative location to the hand, would be better decoded from (parietal/temporal/occipital) ______ lobe.

## *The hierarchy of cortical visual processing*

Neurons sensitive to an object's movement, irrespective of its identity, are more likely to be found in the <u>dorsal</u> or "where" cortical processing stream.

The <u>temporal lobe</u> ventral ("what") processing stream is responsible for recognising, naming, and remembering visually encountered objects based on their identity.

Receptive fields become (more/less) <u>more</u> spatially invariant with increasing depth into the cortical visual processing hierarchy

Would you expect V1 to be (more/less) <u>mode</u> amenable to top-down modulation compared to area IT, deep in the ventral processing stream?

A cortical representation of an object's position, shape, and relative location to the hand, would be better decoded from (parietal/temporal/occipital) <u>parietal</u> lobe.

*end*

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